

Integrity of digital evidences in the investigation process

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Abstract:

Technological preservation of digital evidence from the aspect of integrity is very important for any subsequent handling of data or information contained in said evidence. In practice, it is shown that investigative bodies do not know the technological framework that defines the way of dealing with digital evidence. The occurrence an error during the investigative procedure prevents the necessary subsequent professional actions with an evident lack of completeness of the data and unconfirmed integrity of the extracted data. The paper aims to point out the observed errors in the actions of investigative authorities and define guidelines for further complete and valid actions aimed at delivering data with confirmed integrity.

Keywords:

digital evidence, integrity, encryption, usability, recommendations, standards, procedure

1. Introduction

Data integrity is the concept of maintaining the accuracy, completeness, consistency and reliability of data throughout its entire lifecycle. It involves implementing processes and controls to protect data from loss, corrupting influences and unauthorized access, ensuring that the data is trustworthy and suitable for its intended use. Investigative authorities working with digital evidence must be aware of its vulnerabilities if they hope to maximize its value for justice and accountability [1].

Digital evidence in investigative actions can usually be downloaded only once. After the prepared procedural and organizational prerequisites, it is necessary to carry out the technological procedure of downloading data in the correct way, which will ensure the completeness and usability of data, but inevitably guarantee the integrity of the downloaded data during the entire process of recognition, recording, collecting, transport and storage. In practice, it has been shown that the absence of any technological element of data integrity creates catastrophic problems in the subsequent actions of investigative authorities in the procedural and organizational aspect[2]. The existing ISO standard framework provides clear guidelines for the technological set of actions to be taken to preserve the integrity of retrieved evidence in digital form. Special care should be taken if encrypted data is downloaded, which is not uncommon in the field of global information technology-based data exchange systems.

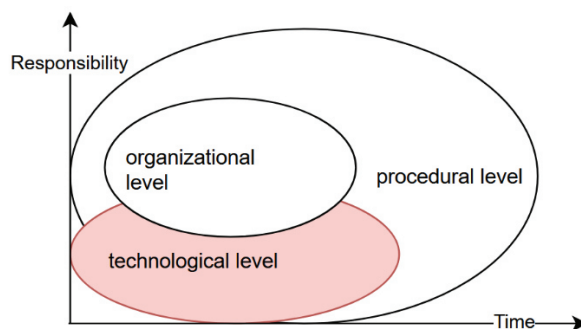


Figure 1: Role of technology in investigation

In trials based on digital data, the importance of the technological framework is crucial from the aspect of presenting data that should, but does not have to be given the status of evidence. The application of technological measures depends on whether the conditions will be met for certain data to be given the status of evidence or whether the entire long-term investigation work was just a waste of time[3].

There are several very good solutions that provide powerful, complementary tools for digital investigations, while enabling robust data extraction and analysis. The software offers solutions based on the latest technologies for detecting and parsing data from seized digital evidence in order to support investigative procedures. These tools must be used with great care, bearing in mind that the investigation is not the end of the operation but only the beginning of a long-term judicial procedure. Practice has shown that engineers who only work as technological experts in the investigation and have no experience in presenting their work in court, do not have a vision for improving their activities that are based only on technologies.

2. Professional rules such as ISO standards

For many years, information technology has not only been an innovative framework defined by the guidelines of university professors, but there are clearly defined rules of the profession that need to be applied. The rules of the profession were first defined as international ISO standards and then adopted as national standards by democratic countries in the modern world. The practice is that international standards may or may not be applied, but that national standards are treated as a clear framework of good practice for application in a particular field.

Global ISO standard ISO/IEC 30121:2015 Information technology - Governance of digital forensic risk framework was adopted back in 2015, while it was adopted as a national standard by the European states from 2015 to 2016: France: NF EN ISO/IEC 30121:2016, Nederland: NEN-EN-ISO/IEC 30121:2016, Belgium: NBN EN ISO/IEC 30121:2016, Slovenia: SIST EN ISO/IEC 30121:2016, Croatia: HRN EN ISO/IEC 30121:2016. EU candidate countries have adopted these standards: Serbia: SRPS EN ISO/IEC 30121:2017, Montenegro: MEST EN ISO/IEC 30121:2017, Bosnia and Hercegovina: BAS EN ISO/IEC 30121:2018. The same practice is in other countries of Europe and the democratic world. ISO/IEC 30121:2015 provides a framework for Governing bodies of organizations (including owners, board members, directors, partners, senior executives, or similar) on the best way to prepare an organization for digital investigations before they occur. This International Standard applies to the development of strategic processes (and decisions) relating to the retention, availability, access, and cost effectiveness of digital evidence disclosure. This International Standard is applicable to all types and sizes of organizations.

Global ISO standard ISO 21043-1:2018 Forensic sciences Part 1: Terms and definitions was adopted in 2018 and in the same year it was given the status of a national standard by numerous countries such as The same practice is in countries in Europe and democratic countries around the world.

The global standard that represents the backbone of digital data processing management is ISO standard ISO 21043-2:2018 Forensic sciences Part 2: Recognition, recording, collecting, transport and storage of items was adopted in 2018. It was adopted as a national standard by the European states: France: NF EN ISO 21043-2:2020, Nederland: NEN-EN-ISO 21043-2:2020, Belgium: NBN EN ISO 21043-2:2020, Slovenia: SIST EN ISO 21043-2:2020, Croatia: HRN EN ISO 21043-2:2020. EU candidate countries have adopted these standards: Serbia: SRPS EN ISO 21043-2:2020, Montenegro: MEST EN ISO 21043-2:2020, Bosnia and Hercegovina: BAS EN ISO 21043-2:2021. This ISO standard specifies requirements for the forensic process focusing on recognition, recording, collection, transport and storage of items of potential forensic value. It includes requirements for the assessment and examination of scenes but is also applicable to activities that occur within the facility. This document also includes quality requirements. ISO standard is not applicable to procedures for the recovery of data from digital storage media which is covered by ISO/IEC 27037. However, the storage medium itself can yield additional items of forensic value (e.g. fingerprints or DNA). Annex D of this Standard shows the applicability of this document to the forensic process.

ISO 21043-3:2025 Forensic sciences Part 3: Analysis. This document specifies requirements and provides recommendations to safeguard the process for the analysis of items of potential forensic value. It includes requirements and recommendations for the selection and application of suitable method(s) for the analysis to meet the needs of the customer and fulfil the request. The requirements are designed to ensure the use of suitable methods, proper controls, qualified personnel and appropriate analytical strategies throughout the forensic analysis of items. This document is applicable to activities conducted by a forensic service provider that occur at the scene and within a facility. This document is applicable to all disciplines of forensic science; however, it is not applicable to the recovery of digital data which is covered by ISO/IEC 27037.

ISO 21043-4:2025 Forensic sciences Part 4: Interpretation. This document specifies requirements and provides recommendations for the interpretation of observations to reach opinions that answer questions that are relevant for decision making in investigations or legal proceedings. This document states requirements that are applicable to all forensic disciplines. This document states requirements that apply when the opinion is based directly on human judgement and when the opinion is based on interpretation through a statistical model. This document is applicable to interpretation that occurs at a scene, within a facility, or within a judicial setting. This document establishes requirements designed to safeguard the process for the interpretation of observations including the use of either statistical models or human judgement, to address alternative propositions based on the questions asked by the customer. Interpretation is not necessary and the requirements of this document do not apply if the observations resulting from the analysis directly answer the relevant question.

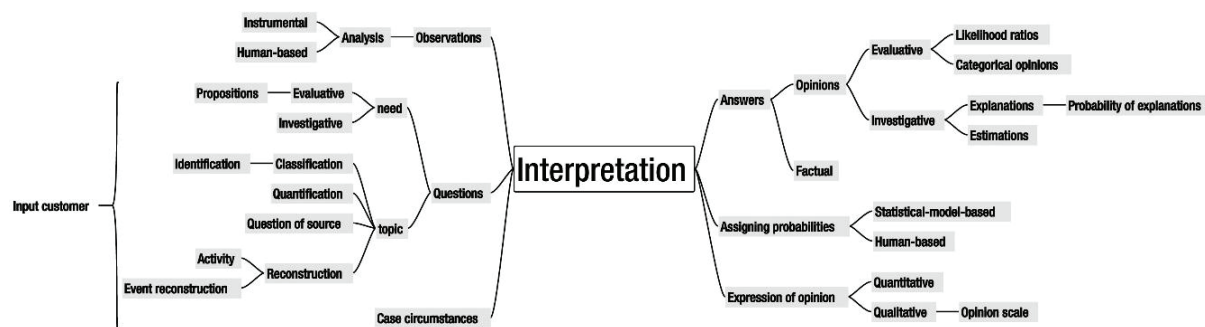


Figure 2: Mind map of forensic interpretation according to ISO 21043 - 4 Interpretation.

Data interpretation is the crucial point that transforms raw data captured into actionable insights, enabling investigators to make data-driven decisions, identify trends for strategic decision-making for the further course of the proceedings. A very important thing at this point is to confirm the identified context of the data. By discovering patterns, understanding complex information and contextualizing data, investigators gain a clear framework for their future strategies towards the courts. Practice has shown that special professional attention must be paid to the difference between Data Analysis and Data Interpretation. Figure 2 shows a mind map that illustrates some of the terminology and structure, input and output of the interpretation process [4]. The standard ends with three informative annexes. The first explains the meaning of the likelihood ratio in the forensic context. The second gives two examples of opinion scales. The third is even more helpful: it essentially takes you through the decision tree in 8 short example cases, with different areas of expertise, types of questions and types of opinion, for both investigative and evaluative interpretation.

ISO 21043-5:2025 Forensic sciences Part 5: Reporting - specifies requirements and provides recommendations for reporting of forensic work. This document is applicable to the reporting of forensic work performed at a scene and within a facility. The requirements facilitate the preparation and issuing of reports that are accurate, clear, transparent, complete, unambiguous, impartial and suitable for their intended use. This document also includes requirements relating to the review of case records and reports, the issuance and control of reports, and requirements related to testimony.

There are certain technological, organizational and procedural rules for retrieving data that investigators hope will become evidence in court. Technological rules, in addition to retrieving data, must include clear and unambiguous context. Retrieving data can most often be done on a device, phone or

computer, internet or communication infrastructure, passive or active data repositories or in the structure of a service provider. Retrieving data that is not encrypted is a simpler process and requires much less documentation than retrieving encrypted data. The process of retrieving encrypted data requires providing the original evidence in encrypted form with all accompanying documentation on the methods and procedures for decrypting it to the level of partial or full readability that will be presented in court. Documentation should ensure that all procedures performed on the initially secured data can be confirmed using reverse engineering.

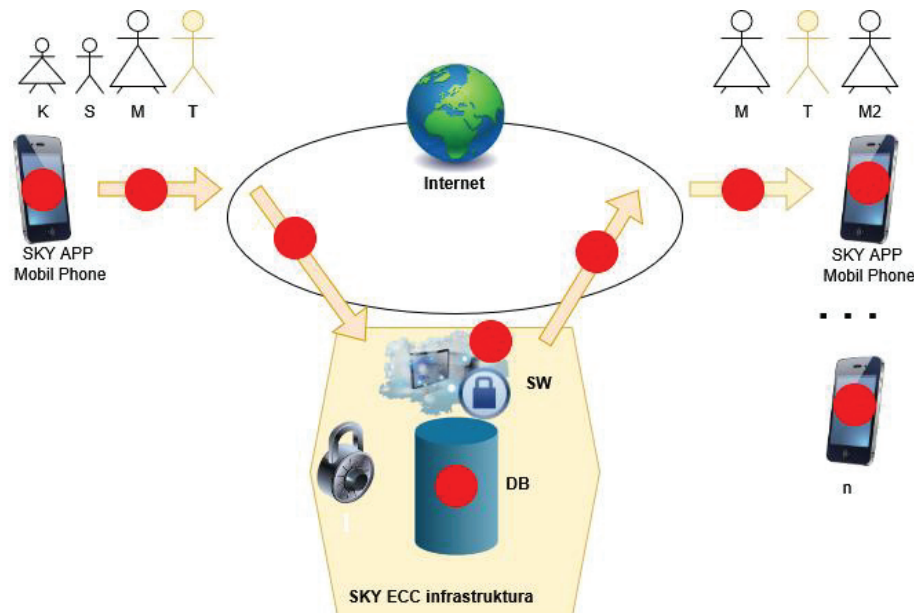


Figure 2: Example of the data interception points

The initially extracted data must be controlled copy of the original data with all necessary documented organizational, procedural and technological actions. All actions on the data must be documented from the initial download of the data by making a controlled copy until the submission of that data to the Court. Each individual action taken must be documented in detail and it is also necessary to adhere to the principle of treating the initially downloaded data as little as possible. Data from the investigation process should not be aggregated with data for which we want to become evidence in court proceedings. The application of artificial intelligence tools is not recommended for data from the investigation[5], and especially not their aggregation with the originally downloaded data. Failure to apply the rules of the profession defined in the reference ISO standards destroys the entire investigation process when such data with an undocumented and undefined technological, organizational and procedural framework appears before the court. In such situations, the court expert for information technology engaged by the Court enters an undefined scope of his work or a scope in which many actions that were initially missed cannot be repeated. The court expert in the proceedings before the court cannot be held responsible for actions missed to preserve the integrity of the data by the investigative authorities.

Special care must be taken when data for investigation purposes is obtained by downloading individual intercepted data outside the infrastructure of the data transmission service provider, rather than downloading data directly in cooperation with and from the database and infrastructure of the service provider[6]. The integrity of data when downloaded directly from the service provider is easier to verify. Downloading individual intercepted data requires documentation for each individual piece of data or set of data, which is a very heavy burden for investigative authorities, but only by applying these principles, the reliability of data can be guaranteed and the offered data have a high chance of obtaining the status of evidence. All of the above applies only to democratic states whose courts respect the legal acts of the state in which the judicial proceedings are conducted.

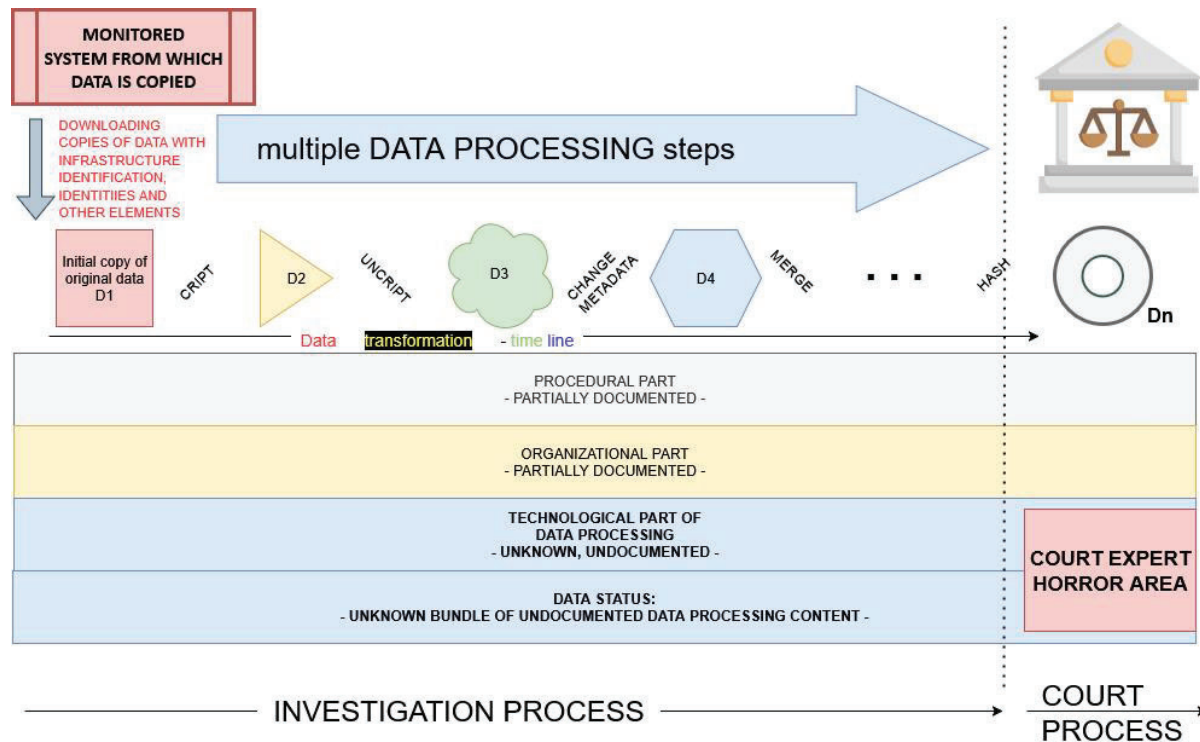


Figure 3: Example of the most common practice - Undocumented data handling

The principles of data integrity in investigative actions are not tied to the institution conducting the investigation, but are tied to each individual piece of data that intends to obtain the status of evidence. The procedures of decryption, hashing, metadata changes and all others must be strictly and completely technologically, organizationally and procedurally documented. Such documentation must accompany each procedure in the investigative process.

In practice, it happens that Courts deliver copies of evidence to the parties to the dispute or the accused in a way that they make undocumented changes to metadata, but such an omission is corrected in a simpler way than an omission that is made during investigative actions.

Within the framework of investigative actions over one large encrypted data exchange system in a court proceeding over the delivered data, the following were identified: intercepted data as well as data processed by artificial intelligence tools - unauthorized data and data from the investigative procedure that were not documented. In addition to all of the above, metadata modification and subsequent undocumented work on the data were identified. Confirming data integrity in such undocumented environment is not possible.

3. Professional rules such as ISO standards

The author's recommendation is that in the process of protecting data integrity when working with digital evidence, the initial principles defined by the ISO 21043 family of standards, as well as additional specific ISO standards, be respected throughout the entire process from Request to Testimony, with the application of the ISO/IEC 30121 standard as well as the application of Specific standards: ISO/IEC 27042, ISO/IEC 27037, ISO/IEC 27050-2, ISO/IEC 27050-3, ISO/IEC 27050-4, and others.

The application of the ISO 21043 standard family is clearly defined: ISO 21043 - 2 Forensic sciences Part 2: Recognition, recording, collecting, transport and storage of items, ISO 21043 - Forensic sciences Part 3: Analysis, ISO 21043 - 4 Forensic sciences Part 4: Interpretation and ISO 21043-5 Forensic sciences Part 5: Reporting.

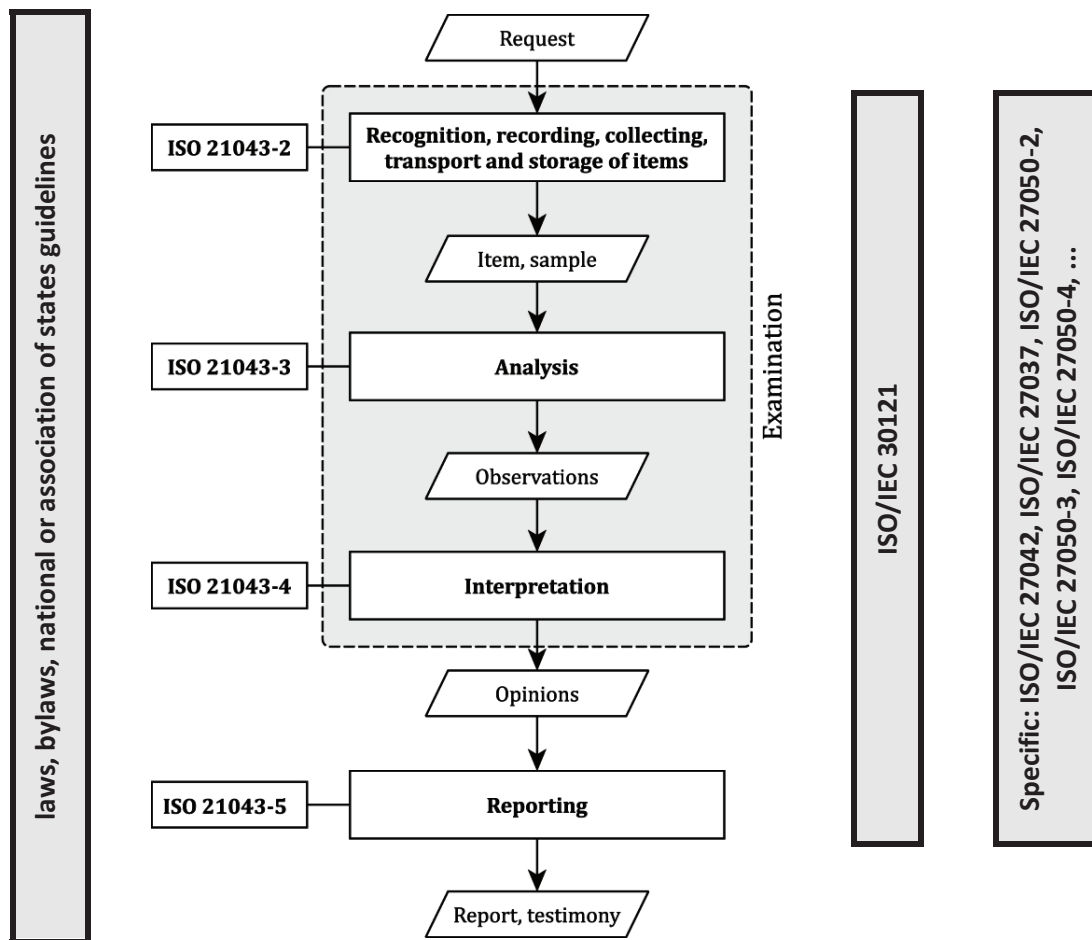


Figure 5: Extended ISO 21043 Model for handling data during investigative actions with the aim of preserving the integrity of evidence

The necessary expansion includes the requirements of the standard for specific needs: ISO/IEC 27042 Information technology - Security techniques - Guidelines for the analysis and interpretation of digital evidence, ISO/IEC 27037 Information technology - Security techniques - Guidelines for identification, collection, acquisition and preservation of digital evidence, ISO/IEC 27050 - 2 Information technology - Electronic discovery Part 2: Guidance for governance and management of electronic discovery, ISO/IEC 27050 - 3 Information technology - Electronic discovery Part 3: Code of practice for electronic discovery, ISO/IEC 27050 - 4 Information technology - Electronic discovery Part 4: Technical readiness and other requirements of the standard according to the specifics of the investigative procedure. The interaction of the requirements of the above standards describes the framework that is minimal for application during the implementation of investigative actions in order to create a valid framework of evidentiary material for court proceedings.

An additional set of extended original data used for investigative purposes should not be submitted to the court in the form of evidence to be processed by a court expert. The recommendation of court experts is to submit to the Court only initially downloaded data in the form of a controlled copy or original data, depending on the specifics of the court proceedings. When submitting controlled copies of files, special attention should be paid to submitting files without modifying the initial metadata. There are a number of ways to preserve the original metadata by copying so that submitted files retain their copy-controlled status.

The Chain of Custody (CoC) is a critical procedural safeguard in digital forensics, ensuring the reliability, credibility, and admissibility of digital evidence in legal proceedings. The Chain of Custody ensuring that evidence collected during an investigation is admissible in court through identified significances: Ensures Evidence Integrity, Establishes Credibility, Supports Legal Admissibility, Prevents

Evidence Contamination, Facilitates Accountability, Preserves Investigation Integrity, Meets Regulatory and Industry Standards [7].

In the Best practice manual for the forensic examination of digital technology[8], Appendix G report requirements section, individual requirements for Forensic Examination of Digital Technology are specifically defined. There is a clear note that the style and content of written evidence must meet the requirements of the criminal justice system for the country of jurisdiction. Section G.1 General Requirements defines 21 requirements that should be included. For cases involving digital evidence (section G.2), the statement or report should also specifically include the: Details of the target data sought; Presence or absence of data found on the various items; and Effects of the examination on the working order of the device. e.g. during the examination the item was (partially) dismantled and the laboratory cannot restore the item to its original state. A peer review (section G.3) of a report should include consideration of the validity of all the critical examination findings and should include ensuring that the correct level of provenance is provided. It should also consider whether the conclusions drawn are justified by the work done and the information available. The review may include an element of independent testing, if circumstances warrant it. A standard check sheet for the technical review covering the following points can be helpful in ensuring that all the relevant issues have been addressed:

- Is the exhibit continuity intact?
- Is the case specific risk assessment available and valid?
- Is there adequate documentation relating to all the items examined?
- Have all the appropriate examinations been carried out?
- Have the relevant procedures and processes been followed?
- Are there complete notes on all the target items?
- Is the statement / report accurate and refer to all items submitted?
- Are the conclusions reached justified and appropriate?
- Was the analysis proportionate?

ENISA(European Union Agency for Network and Information Security) has published a Electronic evidence - a basic guide for First Responders - Good practice material for CERT first responders[9] as a result of its work on good practices for CSIRTs and LEAs in the fight against cybercrime. The document is a guide for first responders, with a particular focus on evidence collection. It aims to complement existing material on the topic of digital forensics and evidence collection, as these are mostly written from a law enforcement perspective. This guide aims to provide guidance on how to handle evidence and the evidence collection process.

Conclusions

In expert circles of digital forensics, the principle of "secure digital chain of evidence" is presented in the complex architecture of systems from which digital evidence is extracted. Basically, it is about "securing" digital evidence on a new complex information and communication system that can be multi-technological and which is based on procedurally, organizationally and technologically organized systems that can provide "digital evidence that can be trusted". Such "trusted digital evidence" is digital evidence that contains a complete number of parameters that inevitably describe the initial (5W1H, who, what, when, where, why, how) but also other technological parameters of the environment, circumstances and state of the infrastructure of the system in which the digital evidence was created. Extraction of incomplete digital evidence cannot be compensated by the institutional importance of the state body, which is, for example, omitted metadata or digital evidence logs.

The framework for handling digital evidence is clear and should be adapted specifically for each investigative procedure and strictly applied in order to ensure the integrity of the data.

The generated Model for handling data during investigative actions given as a model of ISO requirements for the listed ISO standards provides a global framework for implementation. This model is set up in such a way as to set guidelines for investigative bodies on how to handle digital evidence but not shortcuts in dealing with this technological area. Each investigative body in each case in which it processes digital evidence has special specificities that it must identify and select from the presented model. The

specificities of digital evidence are referenced according to the type of digital evidence and the environment from which it is extracted, but certainly also according to the type of equipment and human capacities they possess. The only excuse that is not acceptable in the technological environment of digital evidence is to ignore national ISO standards that have the weight of by-laws. Ignoring standards and not documenting actions that are mandated by ISO standards as good professional practice lead the results of the investigation into a problematic area in the Court. Investigative actions on digital evidence do not end in investigative bodies, but rather in courts with judges who are highly familiar with and apply ISO standards for numerous areas that define the field of information technology.

All of the above represents a clear technological framework that is clearly defined by national ISO standards, but also further clarified by the aforementioned recommendations and studies by formal and expert organizations with the aim of clarifying clearly set instructions on how digital evidence must be handled.

All of the above with the generated Model represents a complete technological framework that is clearly defined by national ISO standards, but also further clarified by the aforementioned recommendations and studies of formal and professional organizations with the aim of clarifying clearly set instructions on how digital evidence must be handled. The Model clarifies that there is no single solution for all problems, just as there is no universal instruction for working with digital evidence, but it provides clear guidelines for experts on how to set the necessary and sufficient level in each individual investigation procedure.

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